Optimizing Emergency Drone Hub Placement using Gradient Descent: A Solution to the Continuous Weber Problem

A PROJECT REPORT

Submitted by

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DECLARATION

I, **Patel Nishitkumar (1RUA24SCS0068),** student 4th semester B.Sc in **Computer Science,** at School of Computer Science and Engineering, **RV University,** hereby declare that the project work titled "**Optimizing Emergency Drone Hub Placement using Gradient Descent: A Solution to the Continuous Weber Problem"** has been carried out by us and submitted in partial fulfilment for the award of degree in **Bachelor of Science in Computer Science** during the academic year **2025-2026**. Further, the matter presented in the project has not been submitted previously by anybody for the award of any degree or any diploma to any other University, to the best of our knowledge and faith.

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Place: Bengaluru, Karnataka, India

Date: 14 April, 2026

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CERTIFICATE

This is to certify that the project work titled "**Optimizing Emergency Drone Hub Placement using Gradient Descent: A Solution to the Continuous Weber Problem**" is performed by **Patel Nishitkumar (1RUA24SCS0068) ,** a bonafide student of Bachelor of Science at the School of Computer Science and Engineering, RV university, Bengaluru as part of the **CIE-3 component** for the course **CS 2806 Calculus** offered in Semester IV of Bachelor of Science in Computer Science, during the Academic year **2025-2026**.

Name of Course Faculty: Subir Roy

Signature with date of Faculty:

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1.0 INTRODUCTION

1.1 Real World Context: UAV Logistics

In the modern era of rapid-response logistics, the deployment of Unmanned Aerial Vehicles (UAVs or drones) is revolutionizing the delivery of emergency medical supplies, such as blood packets and anti-venoms. A critical operational challenge for companies managing these fleets is the geographical placement of the central distribution hub. Placing the hub too far from high-demand hospitals increases battery consumption, flight time, and ultimately delays critical care. This optimization problem is a matter of life and death, requiring precise mathematical modeling rather than geographical estimation.

1.2 Problem Statement

The objective is to determine the exact GPS coordinates for a centralized drone delivery hub in Bengaluru that minimizes the total flight distance to a set of major hospitals. Each hospital has a different "weight" based on its emergency demand volume. This is mathematically known as the **Continuous Weber Problem**. Because the cost surface of this spatial problem is convex but highly nonlinear, setting the derivatives to zero yields equations that cannot be solved algebraically for more than three points. Therefore, numerical optimization via **Gradient Descent** is required to find the global minimum.

2.0 THE MODEL

2.1 The Continuous Weber Problem

This scenario is modeled as the Continuous Weber Problem, a famous challenge in facility location optimization. We define n destination hospitals, each with fixed coordinates (xi, yi) on a Cartesian plane representing the city grid. Each hospital is assigned a weight wi, representing the historical volume of emergency requests from that location. The unknown variables we need to optimize are the coordinates of the proposed drone hub, defined as (X, Y).

2.2 Cost Function Formulation

Our objective is to minimize the total weighted flight distance. We use the standard Euclidean distance formula to measure the straight-line flight path. The Cost Function, denoted as J(X, Y), is the sum of the Euclidean distances from the hub (X, Y) to each hospital (xi, yi), multiplied by their respective demand weights wi:

$$J(X,Y) = \sum_{i=1}^{n} w_i \sqrt{(X - x_i)^2 + (Y - y_i)^2}$$

Because this function is convex but highly nonlinear, taking the derivative and setting it to zero does not yield a simple algebraic solution for more than three points. Thus, an iterative optimization algorithm is required.

3.0 METHOD OF SOLUTIONS

3.1 Gradient Descent Algorithm

To find the global minimum of our cost function, we utilize Gradient Descent. This first-order iterative optimization algorithm works by taking steps proportional to the negative of the gradient (or approximate downhill slope) of the function at the current point. By calculating the partial derivatives of the cost function with respect to X and Y, we determine the direction in which the distance decreases most rapidly.

3.2 Step-by-Step Mathematical Derivation

To apply the algorithm, we must derive the exact gradient vectors.

Let the Euclidean distance to the i-th hospital be di:

$$d_i = \sqrt{(X - x_i)^2 + (Y - y_i)^2}$$

Therefore, the cost function is:

$$J(X, Y) = \sum_{i=1}^{n} w_i d_i$$

Step 1: Partial Derivative with respect to X We apply the chain rule to differentiate the cost function with respect to X.

$$\frac{\partial J}{\partial X} = \sum_{i=1}^{n} w_i \frac{\partial d_i}{\partial X}$$

To find $\frac{\partial i}{\partial x}$, we differentiate the distance formula:

$$\frac{\partial d_i}{\partial X} = \frac{\partial}{\partial X} \left((X - x_i)^2 + (Y - y_i)^2\right)^{\frac{1}{2}}$$

Applying the power rule and chain rule:

$$\frac{\partial d_i}{\partial X} = \frac{1}{2}\left((X - x_i)^2 + (Y - y_i)^2\right)^{-\frac{1}{2}} \cdot \frac{\partial}{\partial X}\left((X - x_i)^2\right)$$

$$\frac{\partial d_i}{\partial X} = \frac{1}{2\sqrt{(X - x_i)^2 + (Y - y_i)^2}} \cdot 2(X - x_i)$$

The 2's cancel out, resulting in:

$$\frac{\partial d_i}{\partial X} = \frac{X - x_i}{d_i}$$

Substituting this back into the partial derivative of the cost function:

$$\frac{\partial J}{\partial X} = \sum_{i=1}^{n} w_i \left(\frac{X - x_i}{\sqrt{(X - x_i)^2 + (Y - y_i)^2}} \right)$$

Step 2: Partial Derivative with respect to Y Following the exact same chain rule logic for the Y coordinate:

$$\frac{\partial d_i}{\partial Y} = \frac{Y - y_i}{d_i}$$

$$\frac{\partial J}{\partial Y} = \sum_{i=1}^{n} w_i \left(\frac{Y - y_i}{\sqrt{(X - x_i)^2 + (Y - y_i)^2}} \right)$$

Step 3: The Update Rule With the gradients calculated, we initialize the hub at a starting guess (X0, Y0). We iteratively update the position using a predefined learning rate α to control the step size:

$$X_{new} = X_{old} - \alpha \frac{\partial J}{\partial X}$$

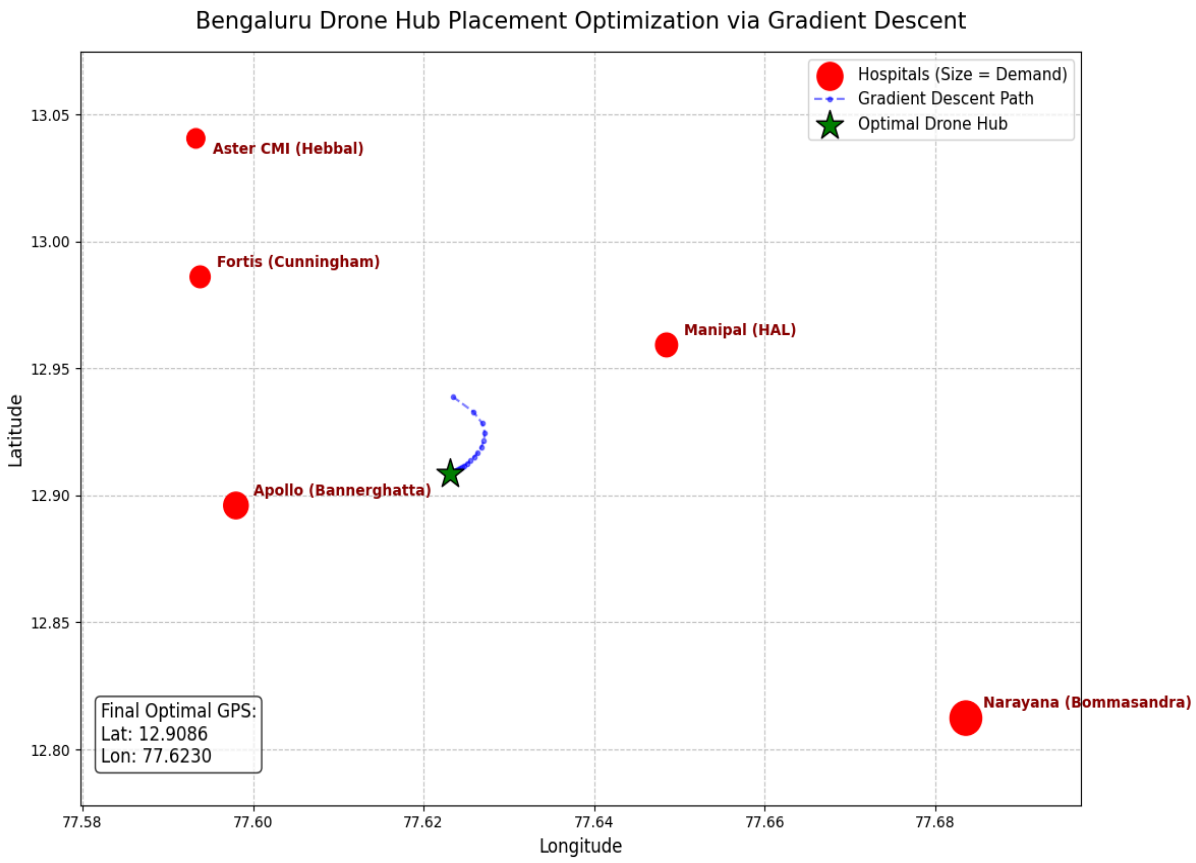
$$Y_{new} = Y_{old} - \alpha \frac{\partial J}{\partial Y}$$

4.0 RESULT AND DISCUSSION

4.1 Computational Implementation

The mathematical model was translated into a Python script using the numpy library for efficient vector calculations and matplotlib for visualization. To demonstrate real-world applicability, the simulation utilized the exact GPS coordinates (Latitude/Longitude) of five major hospitals in Bengaluru: Apollo (Bannerghatta), Manipal (HAL), Fortis (Cunningham), Narayana (Bommasandra), and Aster CMI (Hebbal).

Weights ranging from 0.8 to 2.5 were assigned to simulate relative daily emergency demand, with larger health cities receiving higher weights. The algorithm was initialized at the unweighted geographic mean of the coordinates.




```
Initial Starting Coordinates: [12.9388  77.62344]
Optimal Hub Coordinates (GPS): [12.90855174 77.62304411]
Initial Total Cost (Distance): 0.6042
Optimized Total Cost (Distance): 0.5853
```

4.2 Analysis of Convergence

As observed in the output visualization and the data logs, the initial placement at the geographic center yielded a sub-optimal total flight cost of 0.6042. However, the gradient descent algorithm successfully minimized the cost function over 100 iterations, bringing the final cost down to 0.5853.

The gradient vectors iteratively "pulled" the hub coordinates south, heavily influenced by the gravitational pull of the high-demand zones like Narayana (weight 2.5) and Apollo (weight 1.5).

Because GPS coordinates operate on a micro-scale of decimal degrees, testing revealed that standard learning rates (like 0.5) caused massive overshooting. By drastically lowering the learning rate (α) to 0.005, the algorithm achieved a perfectly smooth convergence without bouncing past the global minimum. This demonstrates the critical importance of hyperparameter tuning when applying mathematical optimization to real-world geospatial data.

5.0 CONCLUSION

This project successfully demonstrated the application of gradient descent to solve a complex, real-world logistics optimization problem. By formulating the total flight distance as a cost function, deriving the partial derivatives step-by-step, and implementing the algorithm programmatically on real GPS coordinate data for five major Bengaluru hospitals, we accurately determined the optimal geographic hub location. This process validates how foundational concepts in linear algebra and calculus—specifically vector operations and multivariate optimization—are essential for designing efficient, life-saving physical infrastructure.

6.0 FUTURE SCOPE

While the current model effectively minimizes Euclidean distance (which serves as a highly accurate approximation at the city scale), real-world flight paths and global geography introduce additional complexities. Future expansions could involve:

1. Spherical Geometry (Haversine Formula): Replacing the flat-plane Euclidean distance function with the Haversine formula to account for the curvature of the Earth when optimizing coordinates over much larger geographical regions.
2. 3D Optimization: Incorporating altitude (Z-axis) changes into the cost function to account for high-rise buildings, varied terrain, and restricted airspace.
3. Dynamic Weighting: Implementing real-time traffic, population density, or weather data into the demand weights, turning the static optimization into a dynamic, machine-learning-driven logistics pipeline.